MITANSHU SHAH

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SKILLS & INTERESTS

- Proficient in C, Java and Python, with a strong foundation in data structures and algorithms (DSA) through problem-solving on LeetCode
- Experience in embedded systems development using Arduino microcontrollers
- Skilled in problem-solving and solution optimization to enhance efficiency and performance
- Enthusiastic about IoT-based innovations and mobile application development

EXPERIENCE

Amee Computer Class Vadodara, GJ
Instructor – Data Structures & Java Programming **August 2025 – November 2025**

- Delivered structured lectures on Java Programming and Data Structures ensuring conceptual clarity and practical application.
- Developed and implemented lesson plans, coding exercises, and assessments to enhance student learning outcomes.
- Provided hands-on guidance in debugging and problem-solving techniques.
- Mentored students to strengthen their programming logic and analytical skills for real-world applications.

ACADEMIC PROJECTS

Automated Smart Composter

- **Embedded Control Software:** Programmed an Arduino microcontroller to fully automate the biological breakdown process, implementing timing algorithms to synchronize mechanical mixing, thermal heating, and exhaust cycles.
- **Hardware-Software Interfacing:** Designed and wired the central control system, seamlessly interfacing the processing unit with a 12V 10A SMPS, digital relay modules, high-torque DC geared motors, and ventilation fans.
- **Logic & Automation:** Implemented robust control logic to manage power distribution and prevent motor stalling under heavy loads, ensuring safe, continuous, and autonomous operation.
- **Systems Integration:** Successfully bridged the gap between software commands and physical actuation by integrating the electronic nervous system into a custom-fabricated mechanical enclosure.

Vahan Sarthak (Android App)

- Created an Android application titled "Vahan Sarthak" using Java programming language. App has been integrated with Firebase Firestore for easy storage and retrieval of data to the users. The application also includes functionalities like login, signup, and moving between pages providing smooth user experience. Presented good backend and frontend development skills on the project, including the efficient use of Firestore for real time database operation.

Smart Tag System V-2

- Developed an Arduino device that scans RFID tags and retrieves associated data from a server, streamlining the challan issuing process and a cloud-based database to enable efficient data retrieval and verification.

EDUCATION

SVIT, Vasad | Vasad, GJ, India
Bachelor of Engineering in Computer Engineering (SPI – 7.74/10) **July 2023 - Present**

Bright School – GSEB Unit | Vadodara, GJ
Stream – Science (Physics, Mathematics, Chemistry, English, Computer) **March 2023**

CERTIFICATIONS

- OOP in Java – UC San Diego (Coursera)
- Python for Data Science, AI and Development by IBM – Coursera
- Technical Support Fundamentals by Google – Coursera
- Operating System and You by Google – Coursera
- The Bits and Bytes of Computer Networking by Google – Coursera

ACHIEVEMENTS

- SIH (Smart India Hackathon) Grand Finalist – 2025
- Secured 2nd position in Vadodara district for Smart Tag system V-1.
- Received 100 days badge on Leetcode and solved 150+ Questions on it.
- Presented project at SSIP initiative and received grant funding for development.